See AI context: https://penguins-eggs.net/llms.txt

Changelog

Release Notes: penguins-eggs v26.7.26 - 2026-07-26

This release brings major enhancements to the Limine bootloader stack, installer auto-detection and confirmation workflows in `sysinstall`, support for Archcraft derivative environments, hardened `makepkg` packaging, LightDM/SDDM autologin PAM fixes, and safe `liveroot` chroot mounting.

🥾 Limine Bootloader & NVRAM Integration

- **NVRAM & EFI Synchronization**: Added UEFI NVRAM registration via `efibootmgr` for Limine installations and synchronized wallpaper configuration across EFI mount paths.
- **Menu Hierarchy & Wallpapers**: Preserved hierarchical tree structures in boot menus and improved splash wallpaper configuration injection.
- **ESP Path & Target Root UUID**: Ensured exact `ESP_PATH` and target partition root UUID injection into Limine configuration files, maintaining consistent `limine.conf` copies across EFI paths.
- **Bootloader Transparency**: Implemented the Principle of Transparency across bootloader configuration modules.

💾 sysinstall & Krill Installer Enhancements

- **Installer Auto-Detection & Launchers**: Added automatic installer detection, `pkexec` desktop launcher support, and an explicit interactive confirmation summary step before proceeding with Krill installation.
- **Dynamic EFI Bootloader Labels**: Dynamically resolved host `DistroID` for EFI bootloader registration to prevent hardcoded `oa-live` labels.
- **Hostname Resolution**: Updated Krill hostname resolution to read default hostname from `/etc/hostname` with fallback to `naked`.

🏹 Archcraft Derivative Support & Packaging Hardening

- **Archcraft Recognition**: Added distro recognition for **Archcraft** under the Arch family definitions (`brain.d/index.yaml`).
- **Makepkg Output Enforcement**: Explicitly forced `PKGDEST` and `PKGEXT` environment settings during `makepkg` execution to align with Alpine packaging conventions, ensuring packages built on Arch derivatives (such as BigLinux) are correctly located in stage.

🔐 Display Manager & PAM Autologin Fixes

- **LightDM Non-Interactive Autologin**: Prepended a service-scoped `pam_permit.so` rule to `lightdm-autologin` PAM stack to eliminate non-interactive login failures ("conversation failed") without touching live user SSH/PAM defaults.
- **SDDM Configuration Guard**: Corrected SDDM autologin setup condition to only write SDDM autologin configurations when SDDM is present.

🐧 Chroot & Mount Propagation Safeguards

- **Liveroot Self-Bind**: Enforced a private bind-mount of `liveroot` onto itself before `chroot` execution with `--make-private` propagation. This ensures tools requiring `/` to be a distinct mountpoint (such as `pacman`'s disk space check) succeed while preventing mount propagation leaks during teardown.
- **Brain Distro Structure**: Split distribution module templates into dedicated `remaster/` and `install/` subdirectories for clearer separation of concerns.

Release Notes: penguins-eggs v26.7.24 - 2026-07-24

This release brings full compatibility and end-to-end remastering support for Arch Linux and its major derivatives (**Arch, EndeavourOS, Garuda Linux and CachyOS**), alongside critical bootloader enhancements for Limine, systemd-boot, and GRUB, installer refinements, and robust initramfs module handling.

🏹 Broad Arch-Family & Derivative Parity (Arch, EndeavourOS, Garuda, CachyOS)

- **Distro Detection**: Extended distribution identification in [coa/pkg/utils/os.go](file:///home/artisan/forge/penguins-eggs/coa/pkg/utils/os.go) to explicitly recognize **CachyOS** and **RebornOS** as members of the Arch family via both `ID` and `ID_LIKE` parameters.
- **Dracut Target Support**: Enabled seamless `dracut` initramfs generation for **EndeavourOS** and **Garuda Linux** within [coa/brain.d/modules/arch-family.bash.tmpl](file:///home/artisan/forge/penguins-eggs/coa/brain.d/modules/arch-family.bash.tmpl), providing explicit target initramfs output paths during installation.
- **Live Storage Drivers**: Added `isofs` and `sr_mod` kernel modules into the live `mkinitcpio` configuration for Arch-family hosts, guaranteeing reliable ISO boot from optical and USB live media.
- **Initramfs Cleanups**: Removed obsolete `fsck` hooks from Arch/Manjaro live `mkinitcpio.conf` templates and ensured kernel version detection uses `uname -r` for accurate initramfs correlation.

🥾 Bootloader Engine Refinements (GRUB, Limine & systemd-boot)

- **Bootloader Selection & Ordering**: Optimized default bootloader prioritization to favor systemd-boot and GRUB, while resolving Limine partition UUID resolution and `fstab` parsing issues.
- **Limine Syntax & Staging**: Fixed Limine configuration syntax (replacing invalid `initramfs_path` with `module_path`, ensuring leading slashes in entry titles), and corrected ESP kernel/initramfs staging.
- **EFI Trampoline Fix**: Corrected copying of the EFI trampoline `grub.cfg` directly into `efi.img` for UEFI bootable media.
- **Symlink Dereferencing**: Ensured `krill` dereferences kernel symlinks (`/boot/vmlinuz-*`) when preparing systemd-boot and Limine boot parameters.
- **Dynamic Distro Labels**: Configured systemd-boot and Limine boot entries to dynamically fetch and display the host's `PRETTY_NAME` from `/etc/os-release`.

🛠️ Krill Installer & Remastering Teardown

- **Simplified Installer Flow**: Streamlined the `krill` TUI installer sequence by removing the manual network configuration screen and defaulting to automatic DHCP configuration.
- **Teardown Error Handling**: Enforced complete unmount sequence during `destroy` teardown operations and properly propagated teardown failures to prevent stale mount points.
- **Alpine Btrfs Hook**: Added `btrfs` initramfs feature generation for Alpine Linux targets running on Btrfs file systems.

Release Notes: penguins-eggs v26.7.18 - 2026-07-18

This release transitions the versioning scheme from semantic versioning (`v0.9.x`) to calendar-based versioning (`vYY.M.D`).

📅 Versioning Scheme Update

- **Transition to CalVer**: Switched the package versioning pattern to calendar-based versioning `vYY.M.D` matching the release date (year, month, day).
- **AUR Compatibility & Versioning**: Adopted this calendar-based format to resolve dependency and update conflicts with the legacy `penguins-eggs` package currently present in the Arch User Repository (AUR). The previous `v0.9.x` versions were treated as older/inferior compared to the old TypeScript-based `penguins-eggs` package versions already in AUR, preventing installation. CalVer solves this upgrade compatibility issue immediately and avoids the need to artificially inflate standard release numbers.

Release Notes: penguins-eggs v0.9.6 - 2026-07-16

This release introduces a new **Go-based Incubator CI** for automated ISO testing, alongside critical bootloader installer fixes, user creation safeguards, vendor branding custom steps, and wardrobe permission enhancements.

🐣 Go-based Incubator CI

- **Native Go Testing**: Replaced the legacy 400-line bash script (`ci/incubator.sh`) with a native Go orchestrator (`incubator-go`) installed on Proxmox.
- **Refined Test Matrix**: Configured automated test runs for `ext4` BIOS, `ext4` UEFI, and `btrfs` UEFI target installs sequentially to prevent server overload.
- **Integrated GHA Reports**: Automatically fetches Markdown test logs from Proxmox and merges them directly into the GitHub Actions step summary using [.github/workflows/incubator-go.yml](file:///home/artisan/forge/penguins-eggs/.github/workflows/incubator-go.yml).

💾 sysinstall & Bootloader Improvements

- **Resilient UEFI Boot**: Tolerate NVRAM registry failures during `grub-install` on UEFI target environments and always populate the `EFI/BOOT/BOOTX64.EFI` fallback inside [coa/brain.d/modules/arch-family.bash.tmpl](file:///home/artisan/forge/penguins-eggs/coa/brain.d/modules/arch-family.bash.tmpl), [coa/brain.d/modules/alpine.bash.tmpl](file:///home/artisan/forge/penguins-eggs/coa/brain.d/modules/alpine.bash.tmpl), and [coa/brain.d/modules/debian.bash.tmpl](file:///home/artisan/forge/penguins-eggs/coa/brain.d/modules/debian.bash.tmpl).
- **Arch UEFI Fallback Fix**: Restored vendor directory and NVRAM registrations for Arch Linux targets by running a standard install first, followed by manual fallback file copying (bypassing the limiting `--removable` flag).
- **systemd-boot Microcode Fix**: Resolved a kernel file corruption issue in systemd-boot configuration templates inside [coa/brain.d/modules/arch-family.bash.tmpl](file:///home/artisan/forge/penguins-eggs/coa/brain.d/modules/arch-family.bash.tmpl) where the microcode initrd line was appended to the kernel line without a newline.

- **Target Network Activation**: Added a dedicated step to enable `NetworkManager` and `systemd-resolved` units inside the chroot in [coa/brain.d/base.yaml.tmpl](file:///home/artisan/forge/penguins-eggs/coa/brain.d/base.yaml.tmpl), ensuring DNS and network work out-of-the-box on the installed target.
- **Mount Bind Fix**: Corrected `extraMounts` options in [coa/pkg/sysinstall/setup/template/mount.conf.tmpl](file:///home/artisan/forge/penguins-eggs/coa/pkg/sysinstall/setup/template/mount.conf.tmpl) back to a YAML list, fixing broken `/dev` and `/run/udev` bind mounts inside Calamares.

⚙️ Calamares & User Management

- **User Creation Collision Fix**: Moved the `removeuser` module right after `unpackfs` in [coa/pkg/assets/calamares_base/settings.conf](file:///home/artisan/forge/penguins-eggs/coa/pkg/assets/calamares_base/settings.conf). This avoids conflicts and failures (exit 9) when creating the new target user with the same username as the live user.
- **Target Failures Monitoring**: The chroot runner script now returns a non-zero exit code if any execution step fails, ensuring installation failures are visible in logs.
- **Autologin Recovery**: Enabled `chroot: true` for the `autologin-gui` module in [coa/brain.d/base.yaml.tmpl](file:///home/artisan/forge/penguins-eggs/coa/brain.d/base.yaml.tmpl) and updated [RunAutologin()](file:///home/artisan/forge/penguins-eggs/coa/pkg/worker/autologin-gui.go#L12) to dynamically handle the custom live username instead of using a hardcoded placeholder.

👔 Wardrobe & tailor Enhancements

- **su Elevation Support**: Enhanced [getWardrobeRoot()](file:///home/artisan/forge/penguins-eggs/coa/pkg/tailor/get-wardrobe-root.go#L24) to resolve the real user's home directory using kernel audit `logname` and `/etc/passwd` scanning, ensuring wardrobe functions work on distros using `su` instead of `sudo`.
- **Early Privilege Check**: Force `coa wardrobe wear` to exit immediately if executed without root privileges, removing redundant internal `sudo` command executions.
- **skel Sync Permissions**: Sincronized `/etc/skel` files using `rsync` with explicit non-root user chown arguments, preventing target home folder assets from being locked by root ownership.

🎨 Custom Vendor Branding

- **Vendor Custom Finish Step**: Support executing a custom vendor `finish.sh` script via [vendorFinishStep()](file:///home/artisan/forge/penguins-eggs/coa/pkg/sysinstall/setup/vendor-finish.go#L25) during the Calamares sequence before the bootloader installation, allowing costumes to customize configuration templates.
- **Branding Asset Overlays**: Allow wardrobes to supply customized logos, slideshows, and `branding.desc` descriptors under the `/etc/penguins-eggs.d/brain.d/assets/calamares/` path.

🛠️ Core Optimizations

- **Native SHA-512 Hashing**: Replaced external `openssl` shell calls with the native Go `sha512_crypt` library in [hashPassword()](file:///home/artisan/forge/penguins-eggs/coa/pkg/planner/hash-password.go#L10) for hashing password inputs during planning.

Release Notes: penguins-eggs v0.9.5 - 2026-07-14

This release introduces **Universal Btrfs Support** across all distributions, along with bootloader customizations, compression optimizations, installer flexibility, and critical robustness enhancements.

💾 Universal Btrfs Support

- **Btrfs GRUB Boot Symlink**: Automatically creates a `/boot` to `/@/boot` symbolic link at the Btrfs partition root level for Arch Linux and Manjaro. This resolves boot failures when GRUB boots from the generic removable UEFI path.
- **Krill Btrfs Detection**: Improved detection logic within the Krill installer to reliably configure Btrfs hooks and initramfs settings when executing inside a chroot environment.
- **Btrfs Batch CI Integration**: Fully automated and verified Btrfs UEFI template cloning and testing within the Incubator Batch CI framework.

🗜️ Compression & Boot Customizations

- **XZ Compression Filter**: Applies the x86 BCJ filter (`-Xbcj x86`) when squashing filesystems using the `xz` algorithm, significantly improving compression ratios on x86/x64 systems.
- **RAM Mode Toggle**: Added configuration options to allow disabling the 'RAM mode' (toram) entry in the generated GRUB and ISOLINUX bootloader menus.
- **Vendor Theme Customization**: Supported custom, vendor-supplied GRUB and ISOLINUX themes and menu labels (`menu-strings.conf`) for complete branding of the boot process.

🛠️ Remastering Robustness

- **Resilient Builds**: Automatically clean up residual mountpoints and staging assets from previous interrupted runs to prevent build contamination.
- **Bypass systemd-firstboot**: Automatically pre-populate essential settings (defaulting `localtime` to UTC, generating a basic locale and hostname) when they are missing in the build source, preventing interactive systemd setup prompts on boot.
- **Idempotent Bootstrapping**: Improved safety and idempotency of the live root bootstrap script.
- **Non-Interactive Tailoring**: Passes `--force-confold` during package installation tasks to guarantee non-interactive executions on Debian-family hosts.

⚙️ sysinstall & Calamares Enhancements

- **Init-Aware machine-id**: Dynamically configures the machine ID setup module based on the host's running init system (supporting both systemd and non-systemd setups).
- **Vendor Users.conf Override**: Allows vendors and costumes to override default Calamares user configurations and password UI requirements.
- **Group Membership Safeguards**: Resolves a bug where the live user could lose their default groups if the remaster was executed from a root shell.
- **Calamares Mount Options Fix**: Fixed a template bug by correcting `extraMounts` options type from list to string.

🏹 Distribution & TUI Adjustments

- **Garuda Linux**: Included full setup support for Garuda Linux in the Arch-family module.

- **Quirinux**: Properly recognized Quirinux as a Debian-family derivative.
- **Sudo-less Wardrobes**: Fixed path resolution for wardrobe configuration detection on systems running without `sudo`.
- **TUI Navigation**: Fixed configuration TUI navigation bugs when skipping hidden fields.

Release Notes: penguins-eggs v0.9.3 - 2026-07-09

This release introduces **Incubator Batch CI**, a new automated testing framework, along with extensive improvements to the `sysinstall` architecture, Krill installer, security defaults, and cross-distribution support.

🐣 Incubator Batch CI (Automated ISO Testing)

- Developed the **Incubator Batch CI** (`ci/incubator.sh` and `.github/workflows/incubator.yml`) to orchestrate automated, end-to-end testing of remastered live ISOs on a Proxmox VE host.
- Deploys and configures a clean QEMU virtual machine for each distribution, booting the generated ISO and executing a fully automated `krill` unattended installation.
- Interacts with the VM via QEMU Guest Agent (`qemu-guest-agent`) to monitor execution, collect installer output, retrieve serial logs, and verify successful boot of the installed system.
- Automatically captures console outputs (`console-SUCCESS-*.log`/`console-FAILED-*.log`) and publishes comprehensive batch reports to GitHub Actions.

🛠️ sysinstall & Krill Installer

- **Filesystem & Partitioning**:
  - Added `--fstype` flag to specify partition filesystem type (e.g., ext4, btrfs) during unattended install.
  - Added automatic partition signature wiping and specified filesystem mount types to prevent Btrfs volume collisions.
  - Integrated dedicated, modular bootloader templates for GRUB, systemd-boot, and Limine on Fedora, openSUSE, Arch, and Manjaro.
  - Full support for Btrfs subvolumes in Krill.
- **Network & Adaptations**:
  - Added Netplan configuration support and fixed DNS formatting in `resolv.conf`.
  - Fallback to active physical network interfaces when the default network route is missing.
  - Improved static IP configuration for Arch Linux (covering dhcpcd, NetworkManager, and systemd-networkd).
- **Hardware & CLI**:
  - Added poweroff option to unattended installations to support automated VM shutdown.
  - Added support for StarFive/Spacemit RISC-V platforms.
  - Dynamic plan execution in the installer using `coa ell`.
  - Configured `calamares` and live-user options directly via the TUI configuration module.

🐧 Distribution Compatibility

- **Fedora**: Implemented a "Kamikaze" service to bypass SELinux blocking during the first post-installation boot, automatically restoring SELinux enforcing mode once the relabel process is complete.
- **openSUSE**: Fixed UEFI/Btrfs bootloader installation and disabled SELinux by default.

- **Devuan**: Added full Devuan support as a Debian-family distribution.
- **Non-systemd Distros**: Enabled tty/console support by forcing `inittab` reloading and hot-reloading configurations.

🔒 Security & Autologin

- Removed `autologin-gui` from default plans to disable automatic GUI login by default, enhancing security.
- Restored desktop autologin wrapper to preserve passwords for CLI and SSH logins.
- Fixed live user sudo permissions by adding explicit passwordless rules for `liveuser`, `linux`, `wheel`, and `sudo` groups in the templates.
- Enabled `nullok` in PAM configuration files during autologin setups to support passwordless credentials.

Release Notes: penguins-eggs v0.9.2 - 2026-06-29

The package officially changed its name from **oa-tools** to **penguins-eggs**.

🚨 BREAKING CHANGE:

A manual uninstallation of the old penguins-eggs package is required prior to installing this release. Ensure your system is clean from the previous version before proceeding.

Alpine Linux — fully stable

Alpine live boot is now fully working. A custom **OA-SIDECAR** is injected into the initramfs during remastering: it intercepts Alpine's standard init after `recovery_shell()`, locates the ISO via `findfs LABEL=OA_LIVE`, mounts the squashfs with an overlayFS layer, and performs `switch_root` into the live system. All six supported distributions (Alpine, Arch, Debian, Fedora, Manjaro, openSUSE) are now stable.

Release Notes: penguins-eggs v0.9.1 - "Functional parity" 2026-06-20

penguins-eggs (oa edition) has reached functional parity with penguins-egg (legacy).

Features

- **Interactive config command** — `coa config` TUI for compression, iso_prefix, password settings
- **Polkit policy** — passwordless installer launch on live/clone/crypted systems
- **Disk space pre-check** — validates free space before remastering starts
- **gshadow/subuid/subgid support** — full user database handling in native C module
- **Comprehensive exclude list** — extended exclusions for cleaner squashfs

Improvements

- **`export iso` uses iso_prefix from config** — no longer builds pattern from distro info when a custom prefix is set
- **`custom.yaml` field-level merge** — individual settings override defaults without replacing the entire block
- **Compression settings from config** — planner injects algorithm/level from config, removed hardcoded values

- **Full mksquashfs command in logs** — replaces summary profile line for better debugging
- **Reordered base.yaml.tmpl** — polkit as its own step, sequential numbering

Code cleanup

- Removed ~370 lines of redundant comments, dead code, and unused struct fields across Go and C
- Added include guard to `logger.h`, fixed double `OE_UID_HUMAN_MIN` define
- Removed tracked `.o` files

Translation

- All Italian user-facing messages translated to English (Go + C)

Full Changelog: https://github.com/pieroproietti/penguins-eggs/compare/v0.9.0...v0.9.1

added ISO size and time in a single success line;

Release Notes: penguins-eggs v0.9.0 - "The Installer" 2026-06-19

Overview

This release introduces **Krill**, the native TUI installer, and completes a deep reorganization of the installation architecture with the `sysinstall` package. It also includes LUKS encryption support, full English internationalization, and significant security hardening.

Krill - TUI Installer

- Native TUI installer with interactive screens for Disk, Users, and Network with real system data
- Reads shared configuration generated by the preparation pipeline
- Full support for both BIOS and UEFI systems
- Unified preparation pipeline shared between Calamares and Krill

sysinstall Architecture

- Complete refactoring: `calamares` → `sysinstall/core`, `sysinstall/calamares`, `sysinstall/krill`
- systemd-boot support
- Prevent sysinstall calamares/krill from running on already installed systems

Security & LUKS

- Full LUKS encryption support with algorithm selection via TUI
- Passphrase passed via stdin, never written to disk
- Fixed security vulnerabilities in C code

CLI & Internationalization

- Translated all Go messages and prompts from Italian to English
- Translated `base.yaml.tmpl`
- CLI cleanup: removed `detect` and `config` commands, hidden internal commands, added `produce` alias

- Hidden `wardrobe` command
- Centralized hardcoded paths in `pkg/config`
- Replaced `--mode` with boolean flags `--clone` and `--crypted`

Remastering & Build

- ISO naming now includes variant (clone/crypted) before the architecture
- Reduced verbose output from dracut/mkinitramfs during remastering
- Clean up residual autologin entries in lightdm.conf
- Locale selection and usage support
- Makefile refactoring
- Universal Calamares configuration

Release Notes: penguins-eggs v0.8.6 - "The Logic Tractor" 2026-06-10

Overview

This release marks a fundamental shift in the development lifecycle of `penguins-eggs`. We have transitioned from a monolithic, script-based approach to a highly orchestrated, multi-layered architecture. By separating high-level planning from mechanical execution, we have transformed the build process into a robust, predictable pipeline.

The Architectural Shift

In developing this version, we embraced a design philosophy rooted in the separation of concerns:

- **The Declarative Layer (JSON/Plan):** We now define *what* the system should achieve through a declarative JSON-based plan. This is the "knowledge" of the build—a high-level map that remains agnostic of the underlying implementation details.
- **The Orchestration Layer (Go):** This acts as the "Logic Tractor." It parses the high-level plan and translates it into specific, deterministic actions. By isolating the logic from the mechanics, we have eliminated the fragility of tightly coupled scripts, ensuring that the control flow remains consistent regardless of the environment.
- **The Mechanical Layer (Bash/C):** This layer handles the "heavy lifting." By delegating the execution to focused, modular Bash scripts, we have replaced complex inline templating with clean, testable, and reliable procedures.

Key Technical Achievements

- **Decoupled Orchestration:** Moving from hardcoded logic to a modular, plan-driven execution model has drastically reduced build failures and increased system observability.
- **Universal Bootloader Logic:** We have standardized the bootloader installation process. By injecting a unified, self-contained installation script (`oa-bootloader.sh`) into the target chroot, we successfully resolved the long-standing cross-compatibility issues between BIOS/Legacy and UEFI systems.
- **Atomic Cleanup:** Through the integration of the final cleanup phase directly into the deployment logic, we have ensured that no live artifacts persist in the installed environment. The system is now as clean as a fresh, official distribution install, with zero manual overhead.

Reflection

Building this system has been an exercise in structural engineering. Just as modern intelligent systems rely on the separation of data, logic, and execution, `penguins-eggs` now follows these fundamental laws. The "Logic Tractor" no longer struggles with the "hoe"—the separation of the orchestrator from the machinery has granted us a level of scalability and stability that was previously unattainable.

This release is not just a collection of bug fixes; it is the foundation for a professional-grade build infrastructure.

Release v0.8.5: The Declarative Evolution 🚀 2026-06-06

This release marks a fundamental architectural milestone for the framework. We have successfully transitioned from a heavily imperative Bash-scripted workflow to a clean, highly robust **declarative YAML orchestration** model.

The engine is now smarter, the codebase is significantly lighter, and the remastering flight plan is easier to read, maintain, and extend.

🌟 Key Architectural Changes

- **The Declarative Engine (`base.yaml.tmpl`)**: The entire live ISO generation process is now orchestrated by a master YAML plan. This provides an Ansible-like, step-by-step clear overview of the remastering and installation hooks.
- **The `oa_ell` Bridge**: Upgraded the internal Go worker (`cmd/ell.go`) to act as a flawless bridge between the C engine and the system. It features a new, indestructible JSON router capable of seamlessly handling tasks like dynamic templating, file copying, and native shell execution.
- **Massive Bash Reduction**: Hundreds of lines of complex, imperative Bash logic have been retired. Core configurations—including bootloader staging (GRUB/ISOLINUX), sudoers setup, trusted desktop launchers, universal autologin, and SVG icon generation—are now handled purely through declarative `ell-actions.tmpl` tasks.

🛠️ Core Improvements & Fixes

- **Unified Squashfs Compression**: Streamlined the root filesystem compression logic (`core_squashfs`). Dropped legacy container-build workarounds in favor of a highly optimized, dynamically injected local profile (defaulting to multi-core `zstd` with automatic fallback configurations).
- **Bulletproof YAML Parsing**: Resolved deep-level scope and Go-template injection edge-cases (`{{- include ... }}`). The `pilot` engine now generates the exact JSON flight plan flawlessly, immune to invisible characters or indentation mismatches.
- **Library Standardization**: Reorganized the template libraries with clear roles and strict 80-column headers (`base.yaml.tmpl`, `scripts.tmpl`, `ell-actions.tmpl`), establishing a clean standard for future open-source contributions.
- **Smarter User Identity Handling**: The internal worker now reliably maps and sanitizes user data, executing native identity injection (e.g., creating the `live` user with specific groups and GID/UID) without dropping to shell scripts.

🎯 What's Next?

With the YAML/Go/C orchestration bridge now fully operational and stable, the framework's foundation is rock-solid. Adding new features or supporting new distributions is now just a matter of adding a few declarative lines to the YAML plan.

[0.8.4] - 2026-06-03

Added

- **New universal `coa tools grub40` command**: Introduced an advanced feature to automate loopback booting of any Linux ISO directly from the host's GRUB menu, eliminating the need for USB flashing during remastering tests.
  - **Ultra-fast inspection via `bsdtar`**: Switched from `xorriso` to native `bsdtar` parsing. The tool now dissects the target ISO in milliseconds, extracting the real paths of Kernel and Initrd from internal configuration files (`grub.cfg`, `isolinux.cfg`).
  - **Dynamic handling for Arch Linux (Kiro/Archiso)**: Resolved the recovery shell boot hang (`device did not show up`) by injecting the GRUB `probe` module. GRUB dynamically calculates the host partition's UUID at runtime, passing it to Archiso via `img_dev` and `img_loop` parameters.
  - **Native Debian/Ubuntu integration**: Seamless support for the `live-boot` engine by automatically mapping the `findiso` boot parameter.
  - **Automation with `--write` / `-w` flag**: Added the ability to directly inject the generated `menuentry` into `/etc/grub.d/40_custom`.
  - **Surgical replacement using markers**: The injection mechanism uses unique identifier comments based on the ISO filename (`# >>> penguins-eggs start...`). This automatically overwrites existing entries when rebuilding the same ISO, preventing file clutter while strictly preserving file execution permissions (`0755`).
  - **Smart host distro detection**: Automatically identifies the host operating system to suggest or run the correct bootloader update command (`update-grub` on Debian/Ubuntu, `grub-mkconfig` on Arch/Fedora).

[0.8.3] - 2026-06-01

Added

- **The Furnace CI**: Fully automated Proxmox/GitHub Actions pipeline for unattended ISO generation.
- Support for automated build matrices across Arch, Debian, Alpine, and Fedora.
- Dynamic IP discovery in CI via targeted ARP sweep (bypassing static MAC/IP configurations).
- `repo add` and `repo rm` commands for cross-distribution repository management.
- Bugfix: Resolved an issue where the ISO dropped to (initramfs) in VirtualBox by implementing the proper .disk/info and .disk/id metadata for Debian live-boot.
- Build: Cleaned up the release version naming in the Makefile by removing the commit hash from the generated tags.
- Added coa tools grub40 command to automatically generate universal GRUB 40_custom entries for booting ISOs via loopback.

Fixed

- Handled missing `/etc/skel` and BusyBox `cp` limitations during the creation of the live user home directory on Alpine Linux.
- Bypassed Linux kernel ping broadcast restrictions during automated CI network discovery.

Changed

- Improved SELinux compatibility during the remastering process for Fedora-based systems.

Deprecated / Beta

- `repo add/rm` commands are currently in Beta for openSUSE and Alpine Linux pending minor compatibility fixes.

🚀 penguins-eggs v0.8.2: Packaging fixes, dynamic exclusions & Config rollout

I believe these new features will be highly interesting for many users: the ability to change the live user password and customize the compression algorithm, as well as easily add or remove custom exclusions during the remastering process.

- **feat(config):** officially published the configuration file (`/etc/penguins-eggs.d/custom.yaml`, previously omitted by mistake). Users can now easily customize system-wide settings, including the live user password, as well as mksquashfs compression algorithms and parameters.
- **feat(engine):** revamped `/etc/penguins-eggs.d/custom.exclude.list` with robust parsing (elegantly ignoring comments and blank lines) and populated it with default exclusions for heavy container engines (Docker, Podman, LXC/LXD) and Snap. Users can easily disable these defaults or expand the list at will to keep their generated ISOs perfectly lean.

🚀 penguins-eggs v0.8.0: The Architectural Leap & Expanded Horizons

This is a fundamental release for `penguins-eggs`, introducing a rock-solid template architecture and extended support for new distributions.

✨ Key Features & Improvements

- **Bulletproof Template Architecture:** Introduced strict `core_` and `hook_` nomenclature for all templates. This eliminates silent "shadowing" conflicts and provides a highly scalable foundation (ready for the upcoming Bianbu OS / RISC-V porting).
- **Alpine "Sidecar" Parachute:** Implemented a native recovery system in the initramfs for Alpine Linux, ensuring stable OverlayFS mounting and virtual filesystem relocation right before OpenRC boot.
- **openSUSE Support:** Added full support for openSUSE, bringing `penguins-eggs` on par with the historical parent distributions supported by `penguins-eggs`.
- **CLI & UX Enhancements (Cobra):** * Native shell auto-completion (Bash/Zsh/Fish) dynamically generated during `.deb` packaging for both `coa` and the legacy `eggs` alias.
    - New `coa export log` command to instantly extract debug logs to a remote host via SSH.
- **Semantic Polish:** Renamed the destructive command to `eggs destroy` to align with standard DevOps terminology.

🐧 penguins-eggs v0.7.9 - The CI/CD & Architecture Release

Evolution Architecture

- **Decoupled Engine & Pilot:** Completely refactored the Go architecture to eliminate import cycles. Environmental awareness (like detecting GitHub Actions) is now handled natively by the `cmd` package

(Command Bridge), leaving `engine` and `pilot` strictly agnostic and modular.
- **Dynamic Template Injection:** The `TemplateContext` now dynamically receives environmental flags to adapt script generation on the fly without logic duplication.

CI/CD & Automation Breakthrough

- **GitHub Actions Turbo Mode:** Introduced automatic CI detection. When running on GitHub Actions, the system auto-injects a "Turbo" profile for `mksquashfs`, applying aggressive zstd compression and smart exclusions to slash ISO size (down to 6.4GB) and drastically reduce build times.

🐧 penguins-eggs v0.7.8 -

- **Smart Desktop Management**: Installation links are now dynamically handled and automatically removed after a successful installation, keeping the target `/etc/skel` clean.

Evolution Architecture

- **Modular Template Engine**: Decoupled YAML orchestration from Bash logic using a new "Helm-style" system.
- **Enhanced Bash Isolation**: Moved all scripts to dedicated `.bash.tmpl` files with zero-margin indentation for better maintainability.

Technical Improvements

- **Cross-Distro Stability**: Verified seamless support for Debian, Arch, Fedora, and Manjaro using the new architecture.
- **Pilot Logic**: Added `include` and `indent` functions to the Go engine for perfect YAML syntax generation.

Release v0.7.7

Core Engine & Remastering

- **Fixed SquashFS Exclusions:** Corrected wildcard expansion rules (`.*`) in `mksquashfs` exclude lists that previously caused the unintended removal of vital directories (like `/mnt`, `/media`, and `/root`) during the live filesystem compression.

Live System & Templates

- **Universal Autologin:** Replaced distro-specific workarounds with a unified, robust LightDM autologin bypass. It successfully handles PAM restrictions and locked passwords across Debian, Arch Linux, and Fedora live sessions.
- **Trusted Desktop Launcher:** Implemented an automated background script ("Secret Agent") that dynamically applies GIO metadata and checksums to authorize the Calamares installer icon across all major Desktop Environments (XFCE, GNOME, KDE, Cinnamon, etc.) without security warnings.
- **Dynamic 3D SVG Icon:** The live system now dynamically generates a high-quality, 3D-styled SVG egg icon directly from the YAML template.

Calamares Installer

- **Slideshow Redesign:** Overhauled the `show.qml` presentation with a new "Arctic Gold" aesthetic, featuring improved typography and high-contrast text outlines for perfect readability over snow/bright backgrounds.
- **Responsive Image Scaling:** Introduced a dynamic mathematical container in QML. Slideshow images now scale perfectly to fit the installer window while locking the text inside the image boundaries, eliminating text overflow and blank borders.

Release v0.7.6: New Template Architecture** - 2026-05-10

This release marks a fundamental structural shift for `penguins-eggs`: the transition to a modular build system.

We have separated the system logic using templates:

- **Universal framework:** Actions common to all systems are now centralized in `brain.d/base.yaml.tmpl`.
- **Specific modules:** Code unique to each distribution (Debian, Arch, Fedora, Manjaro) has been isolated within `brain.d/modules/`.

This approach drastically reduces code complexity compared to previous versions and simplifies readability. Beyond stabilizing our current work, this architectural cleanup provides the structural foundation necessary to explore new developments and support scenarios that were previously out of reach.

[0.7.5] - 2026-05-03

Added

- **Universal Boot Branding**: Integrated support for custom splash screens (`splash.png`) and Unicode fonts (`unicode.pf2`) for both BIOS (Isolinux) and UEFI (GRUB).
- **Enhanced GRUB Layout**: Added a dynamic title and visual spacing to the GRUB menu for a professional look.
- **Dynamic OS Detection**: Boot menus now automatically display the distribution name (via `$PRETTY_NAME`).
- **Comprehensive Shell Support**: Full completions for Bash, Zsh, and Fish, including automatic 'eggs' alias registration.
- **Fedora UI Fixes**: Added `google-noto-emoji-fonts` as a dependency in the RPM builder to ensure correct symbol rendering in the terminal.

Fixed

- **Fedora UEFI Visibility**: Added `efi_gop` and `efi_uga` modules to GRUB configuration to fix splash screen issues in UEFI mode.
- **Isolinux Compatibility**: Standardized the `APPEND` syntax for boot parameters, ensuring a successful boot on Fedora and other non-Debian systems even when using Debian bootloaders.
- **RPM Asset Packaging**: Updated the Go builder to correctly package and deploy branding assets into `/etc/penguins-eggs.d/brain.d/assets/`.

Technical Notes

- The "oa" dialect is now fully established for egg-based eggs-bananas remastering.

- All configurations are now centralized in the "brain" directory for better portability.

Release 0.7.4: The Great Refactoring

Fedora is now aligned to the others distros: arch, debian and manjaro.

This release marks a turning point for **penguins-eggs**. We have moved beyond chasing the specific quirks of individual distributions to build a universal, fluid, and frictionless infrastructure.

🏗️ Architectural Evolution

The core has been surgically redesigned to separate concerns between the "Brain" and the "Muscle":

- **The Orchestrator (Go)**: Now acts as the project's intelligence. It parses profiles, manages logic, and generates a perfect "flight plan" (**oa-plan.json**). Its predictive logic prevents failures before they even begin.
- **The Engine (C)**: The `oa` binary is the operational arm. It consumes the JSON plan and interacts directly with the OS with the speed and precision of C, handling mounts, chroots, and identity injection in a secure, isolated environment.
- **Unified Dialect**: We have standardized how tasks are defined. The keys `action`, `description`, and `run_command` now drive every operation, making profiles significantly more readable and maintainable.

🔮 Coming Soon: Profile Simplification

While this release solidifies the backend, we are already looking ahead. **The next step is a massive simplification of the profile structure.** By implementing **Go Templates**, we will drastically reduce boilerplate code, allowing a single profile to adapt dynamically to different environments and architectures without manual duplication.

🛠️ Technical Highlights

Feature	Description
Cross-Distro Validated	Full support for Debian (Trixie), Arch, Fedora, and Manjaro.
Robust Chroot Environment	Automatic `PATH` handling and *UsrMerge* link replication to ensure post-install success.
JSON-Driven Execution	Complete decoupling of task definition from operational execution.
Advanced Emergency Cleanup	Improved "nest" clearing to ensure the system returns to a clean state after every run.

[0.7.3] - 2026-05-01

Core Improvements & Fixes

Full compatibility achieved with Linux Mint, the Ubuntu family, and its derivatives.

[0.7.2] - 2026-05-01

Core Improvements & Fixes

- **Standardized Sudo Privileges**: Implemented a unified "Arch-style" sudo configuration for the live user, ensuring seamless operation of **oa** tools across Arch, Debian, and Manjaro bases.
- **Enhanced Build Cleanup**: Integrated a "tabula rasa" policy during the remastering process to automatically remove legacy or malformed configuration files (e.g., `00_artisan`) that previously caused syntax errors in the sudoers directory.
- **Post-Installation Optimization**: Added an automated cleanup routine to the **FINALIZE** stage. This ensures that the installed system is stripped of live-environment "ghost" files, leaving only the necessary installer-generated permissions.
- **Permission Hardening**: Enforced strict `0440` permissions for generated sudoers files to comply with security standards and prevent them from being ignored by the system.

This release marks a significant step in the **eggs-bananas** philosophy, delivering a cleaner, more professional transition from the live environment to the final installation.

[0.7.1] - 2026-04-30

"The Streamlined Artisan" Update

This minor release focuses on code purification and structural robustness, following our core "Eggs & Bananas" philosophy. We've removed redundant configuration files and refined the distro-agnostic engine for better performance and reliability.

Added

- **Smart Directory Management**: The Arch Linux and Manjaro builders now correctly install the `brain.d` logic directory into `/etc/penguins-eggs.d/`, ensuring agnostic mapping is available immediately after installation.
- **Enhanced Distro Detection**: Integration with system `ID_LIKE` metadata allows for seamless recognition of derivative distributions without external mapping files.

Changed

- **Code Purification**: Removed the `DistroUniqueID` field and the `derivatives.yaml` dependency. The system now relies on pure `/etc/os-release` data, making the codebase leaner and easier to maintain.
- **Refactored NewDistro**: Optimized the recognition logic to prioritize system-provided identity over manual overrides.

Fixed

- Fixed a missing directory copy in the Arch Linux `PKGBUILD` that prevented `coa` from finding its core logic.
- Improved error handling in the `distro` package: the system now enters a "Generic" mode instead of exiting when encountering an unmapped distribution.

Release Note: oa 0.7.1 - Simplicity is the Ultimate Sophistication

With **oa 0.7.1**, we continue to strip away the unnecessary. By moving the distribution mapping directly into the core logic and leveraging system metadata, we've eliminated the need for external configuration files.

This release ensures that your "Artisan" environment on Arch Linux is configured perfectly from the first `makepkg`. It’s faster, cleaner, and more robust.

The artisan's tool just got sharper.

[0.7.0] - 2026-04-29

"The Agnostic Artisan" Release

This release implements the **"Eggs & Bananas"** philosophy: reducing maintenance costs by 90% while retaining the power of the original wardrobe. The tailor logic is now lighter, faster, and ready for a multi-distro world.

Added

- **Distro Agnosticism**: `oa` now detects the running distribution and adapts its "tailoring" strategy.
- **AI-Assisted Tailoring**: For non-Debian systems (like Arch Linux), `oa` now generates a comprehensive `AIPrompt.txt` directly in the user's Home directory.
- **Hardware-Aware Prompts**: The AI prompt now includes `lspci` data (VGA/3D controllers) and available `xsessions` to help the AI suggest the correct drivers and display manager configurations.
- **Smart APT Filtering**: On Debian-based systems, `oa` now verifies package existence via `apt-cache` before attempting installation, preventing a single missing package from breaking the entire process.
- **Robust Path Resolution**: Post-installation scripts (using `../../scripts/` syntax) are now automatically resolved to absolute paths and granted execution permissions.
- **Persistent Logging**: Added `/var/log/coa-tailor.log` for background auditing and improved user-facing logs.

Changed

- **Flat YAML Structure**: Deprecated the nested `sequence`/`finalize` blocks in favor of a flat, clean `index.yaml` at the root of each costume/accessory.
- **Simplified Wardrobe**: Removed legacy repository management from the wardrobe logic to favor system-native tools or AI guidance.
- **Refactored Tailor Engine**: The `wear` logic is now 90% simpler to maintain while remaining 90% as convenient as the original implementation.

Fixed

- Fixed script execution errors where relative paths were not found by the shell.
- Fixed `sudo` environment issues: `AIPrompt.txt` and synced home files now correctly belong to the real user even when run with root privileges.

Release Note: oa 0.7.0 - The Agnostic Artisan

We are proud to announce the release of **oa 0.7.0**.

In the spirit of pragmatism, we have stripped away the complexity that made cross-distro support difficult. By embracing the "Eggs & Bananas" approach, we've created a "Tailor" that isn't just a package installer, but an intelligent assistant.

Whether you are on a "naked" Debian or a fresh Arch Linux installation, `oa` helps you dress your system with your favorite costumes. If it can't install a package automatically, it provides you with a "Medical Record" of your system (`AIPrompt.txt`) to hand over to an AI assistant, ensuring you get the perfect configuration for your specific hardware (including tricky ones like QXL or Nvidia).

It's time to sow the seeds of a truly universal Linux customizer. ` Ho sistemato il Changelog includendo tutti i punti chiave discussi. Buona semina per questa versione 0.7.0! 🐧🚀

[0.6.5] - 2026-04-15

🚀 Added

- **"eggs" Alias Support**: Implemented dual binary identity. The package now installs symlinks and shell completion patches (Bash/Zsh/Fish) to allow using the `eggs` command as an alias for `coa`.
- **Standardized Naming Engine**: Isolated naming logic into `naming.go`. ISOs now follow the format: `egg-of_[distro]-[codename]-[hostname]_[arch]_[timestamp].iso`.

🛠️ Changed

- **CLI Semantic Refactoring**: Renamed core commands for clarity: `produce` is now **`remaster`** and `krill` is now **`sysinstall`**.
- **Build System Decoupling**: Split the monolithic `build.go` into distribution-specific handlers (`build_debian.go`, `build_arch.go`, `build_fedora.go`) for granular dependency and metadata management.
- **Universal Boot Extraction**: Refactored boot file extraction to natively support Arch, Debian, and Fedora structures during the `remaster` phase.
- **Arch Linux Initrd Refactor**: Optimized ramdisk generation on Arch via deeper `mkinitcpio` integration.

🐛 Fixed

- **Fedora 43 Boot (BLS/EXT4)**: Resolved "grub rescue" issues by forcing the deactivation of *Boot Loader Specification* (`GRUB_ENABLE_BLSCFG=false`) and applying EXT4 compatibility flags (`^metadata_csum_seed`, `^orphan_file`) during formatting.
- **Installer Hang (Sync & Cleanup)**: Fixed a deadlock during the final Krill phase on Fedora by making `/var/lib/live` cleanup selective for Debian and introducing a forced `sync` before unmounting partitions.
- **Chroot Pathing**: Fixed path resolution errors during `oa_shell` execution in chroot environments, ensuring correct injection of hostname and sudoers.

Packaging

- **RPM Dependency Mapping**: Corrected Fedora RPM metadata by mapping `sgdisk` requirements to the `gdisk` package.

- **Symlink Integration**: All native packages now include symlinks for binaries, man pages, and shell completions for the `eggs` alias.

[0.6.1] - 2026-04-10

Fixed

- **Arch Linux Live Boot**: Resolved a critical issue causing Kernel Panic on boot. The ramdisk generation now correctly includes the `archiso` hook.
- **Initramfs Chroot Enforcement**: Fixed a bug where `mkinitcpio` and `mkinitramfs` were reading host system configurations instead of the target `liveroot` configurations. Commands are now explicitly forced to use `-c` or `-d` flags pointing to the chroot.
- **Bootloaders Tarball Extraction**: Restored the directory flattening logic (`parts[1:]`) in the `extractTarGz` utility to prevent nested root folders when downloading from GitHub releases.
- **Engine Logs**: Suppressed false-positive `cp` error outputs (`stderr`) in the C engine (`lay_livestruct.c`) during fallback kernel extraction attempts, resulting in cleaner build logs.

Changed

- **Universal Bootloaders Path**: Unified the `BootloadersPath` assignment across all distribution families (Debian, Arch, Fedora, openSUSE). All environments now consistently use the universal `BootloaderRoot` (`/tmp/coa/bootloaders`).

Packaging

- **Dependency Fixes (Arch)**: Added `archiso`, `dosfstools`, and `mtools` as mandatory dependencies in the `PKGBUILD` to ensure successful live ISO generation and UEFI support.
- **Dependency Fixes (Debian)**: Added `dosfstools` and `mtools` to the Debian `control` file.
- **Legacy Conflict**: Added an explicit conflict with the `penguins-eggs` package in both Debian and Arch builds to prevent overlap and ensure a clean `oa` ecosystem.

[0.6.0] - 2026-04-09

🚀 Added

- **Universal Agnostic Partitioning**: The C engine now always initializes target disks with a 3-partition GPT table (`BIOSBOOT`, `EFI`, `ROOT`), ensuring universal compatibility and disk portability regardless of the host firmware.
- **Legacy BIOS Support**: Implemented the `hatch_bios.c` action to execute `grub-install --target=i386-pc` directly onto the physical disk.
- **Dynamic Routing (Krill)**: The Go orchestrator dynamically detects the presence of `/sys/firmware/efi` and injects the correct bootloader installation action (`hatch_uefi` or `hatch_bios`) into the JSON flight plan.
- **Dynamic Squashfs Detection**: The orchestrator (The Mind) automatically locates the pristine `filesystem.squashfs` image based on standard live distro mount points (e.g., `/run/live/medium/...`) and passes the exact path to the C engine.

🛠️ Changed

- **Pristine Install Architecture (Unpack)**: Replaced `rsync` with `unsquashfs` in `hatch_unpack.c`. The installer no longer clones the actively running (and potentially polluted) Live system, but extracts the pure factory image for a clean, pristine installation.
- **Unpack Progress Bar**: Restored the standard output of `unsquashfs` during the hatch phase to display the native progress bar to the user.
- **EFI Image Size**: Increased the `efi.img` file size for hybrid ISO generation from 4MB to 10MB to safely accommodate larger modern bootloaders.
- **FAT16 Compliance**: Forced the EFI image for hybrid boot into the `-F 16` format to satisfy strict validation checks by modern UEFI firmwares (such as OVMF on Proxmox).

🐛 Fixed

- **Hybrid ISO Boot**: Fixed UEFI boot failures on the generated ISO. Added the `-append_partition 2 0xef` flag to `xorriso` to physically inject the EFI partition into the ISO's GPT table, making it universally bootable as both a CD-ROM and a USB drive (Raw Disk).
- **Removable Media Fallback**: Corrected the UEFI bootloader path inside the ISO to strictly comply with the removable media standard (`/EFI/BOOT/bootx64.efi`).
- **Partition Syncing**: Fixed a bug in `hatch_partition.c` where the BIOSBOOT partition (2MB) creation was skipped, which caused an index shift and subsequent formatting failures.

[0.5.1] - 2026-04-08

[coa] Orchestrator (The Mind)

- **New Build System**: Added `coa build` to automate C/Go compilation and native packaging (`.deb`, `PKGBUILD`).
- **Dynamic Exclusions**: Now generates runtime exclusion lists for SquashFS, moving logic away from the C engine.
- **Auto-Docs**: Integrated `coa docs` to generate man pages and shell completions (Bash, Zsh, Fish).
- **Version Controller**: Established as the Single Source of Truth for the entire project versioning.

[oa] Engine (The Body)

- **Decoupled Logic**: Removed hardcoded exclusion arrays in `lay_squash.c` in favor of dynamic `-ef` flags.
- **Version Injection**: The engine now receives its version string during compilation via Makefile macros.
- **Streamlined Execution**: Focused the C core on pure system actions, delegating policy-making to the orchestrator.

[General]

- **Git Hygiene**: Optimized `.gitignore` to keep the repository clean from build artifacts.
- **Single Source of Truth**: Centralized versioning in `build.go`.

[v0.5.0] - 2026-04-02

🚀 Added

- **Smart Surgical Unmount**: `action_cleanup` now dynamically reads `/proc/mounts` to perform an inside-out unmount (`MNT_DETACH`). It specifically targets only `liveroot` and `.overlay` branches, eliminating "zombie mounts" safely without affecting user network shares or external drives.
- **Absolute Path Support for ISO**: `action_iso` now detects if the `output_iso` parameter is an absolute path (starts with `/`). This allows the engine to write the final ISO directly to external mounts (like a remote NAS or USB drive) bypassing local storage entirely, removing the need for `.mnt` hidden folders or symlinks.
- **Agnostic Bootloader Injection**: Introduced the `bootloaders_path` JSON parameter. `action_uefi` and `action_isolinux` can now dynamically pull bootloader binaries (like Debian's `grubx64.efi` and `isolinux.bin`) from an external path provided by the orchestrator, enabling universal booting for non-Debian host systems.
- **Hybrid UEFI/BIOS ISO Master**: `action_iso` now automatically detects the presence of UEFI payloads and dynamically generates a 4MB FAT `efi.img` (via `dd`, `mkfs.vfat`, and loop mount) to inject into `xorriso` using the `-eltorito-alt-boot` flag.
- **UEFI Trampoline Configuration**: Added a secondary `grub.cfg` inside the `EFI/BOOT/` directory to act as a trampoline. This redirects the firmware to the main `/boot/grub/grub.cfg` on the ISO9660 filesystem, completely avoiding the dreaded `grub rescue>` prompt.

🛠️ Changed

- **Single Responsibility Principle Enforced**: Completely removed legacy, hardcoded cleanup routines from `action_prepare.c`. Now `prepare` only mounts, and `cleanup` only unmounts.
- **Plan JSONs Updated**: All JSON templates (`plan-standard.json`, `plan-clone.json`, `plan-crypted.json`) have been updated to include the `bootloaders_path` key and the correct modern modular actions.
- **Code Refactoring**: Removed the legacy monolithic `action_remaster` logic completely, distributing its responsibilities natively into `action_livestruct`, `action_isolinux`, and `action_uefi`.

🐛 Fixed

- **Missing GRUB Modules**: Fixed an issue in `action_uefi` where `*.mod` and `*.lst` files were not copied to the `x86_64-efi` directory, causing boot failures.

📚 Documented

- **Universal Strategy Manifesto**: Created `docs/UNIVERSAL_STRATEGY.md` outlining the 4 pillars of the *penguins-eggs* ecosystem (Debian Passepartout, Yocto-style identity, Initramfs abstraction, and the Orchestrator role).
- **Architecture Guide Update**: Added details about the `tmpfs` Anti-Inception shield and dynamic `/home` handling to `docs/ARCHITECTURE.md`.
- **Actions Reference**: Completely overhauled `docs/ACTIONS.md` to reflect the new C engine modules (added `cleanup`, `crypted`, `scan`, `suspend`).
- **README and Roadmap**: Cleaned up `README.md`, moved future goals to a dedicated `docs/ROADMAP.md`, and added links to the philosophical architecture.

[0.2.0] - 2026-04-01

Added

- **Anti-Recursion Shield (Inception Fix)**: Implemented a global `tmpfs` mask in `action_prepare.c` to hide the working directory (`LiveRoot`) from `mksquashfs` and `nftw`. This definitively prevents infinite filesystem loops when the workspace is located inside a bind-mounted host directory (like `/home`).
- **Native Group Injection**: Added the `yocto_add_user_to_groups` helper in `oa-yocto.c` to natively append the live user to secondary groups (e.g., `sudo`, `cdrom`) directly into `/etc/group`, completely bypassing host binaries.
- **Skeleton Population**: `action_users` now correctly populates the live user's home directory by copying hidden configuration files from `/etc/skel` and applying recursive ownership.

Changed

- **Smart `/home` Handling**: Refactored `action_prepare.c` to handle the `/home` directory dynamically based on the execution `mode`. It is now mounted read-only for `clone` and `crypted` modes, but created as an empty directory for `standard` mode to host the newly injected live user.
- **Cleanup Fortification**: Updated `action_cleanup` to safely unmount the new anti-recursion masks and the `/home` directory using `umount2(..., MNT_DETACH)`.

Fixed

- **Missing Live Home in ISO**: Fixed a bug in `action_squash.c` where the `home/*` exclusion was aggressively deleting the freshly created live user's home during the `mksquashfs` compression in `standard` mode.

[0.1.0] - 2026-03-30

Added

- contest.sh create a context.XXX.txt to restore GEMINI contest;
- **Modular Architecture**: Decoupled core logic into `src/actions/`, making the project significantly more scalable and maintainable.
- **Execution Engine**: Implemented the `execute_verb` dispatch system in `main.c` to process the `plan.json` workflow.
- TODO: **Parameter Inheritance**: Added dual-pointer support (`cJSON *root` and `cJSON *task`) allowing actions to access both Global settings and Local task overrides.
- **Dynamic Initrd Action**: Support for command templates using `{{out}}` and `{{ver}}` placeholders for flexible initramfs generation.
- TODO: **System User Discovery**: Added `action_users` to scan for "human" users (UID >= 1000) using the POSIX `getpwent()` API.

Changed

- **Buffer Hardening**: Introduced `PATH_SAFE` (8192) and `CMD_MAX` (32768) constants in `oa.h` to ensure safety during complex `system()` command construction.
- **Centralized Definitions**: Consolidated all system headers and function prototypes into a single master header (`include/oa.h`).
- **Mount Fortification**: Refactored `action_prepare` to use `MS_PRIVATE` bind mounts and improved OverlayFS directory structures.

Fixed

- **Warning Purgatory**: Resolved all GCC warnings related to `-Wformat-truncation` and `-Wunused-parameter`, resulting in a "Zen" build.
- **Logic Correction**: Fixed the Megabyte calculation in `action_scan.c` by replacing the incorrect `PATH_MAX` divisor with a proper `1024*1024` calculation.
- **JSON Key Sincronization**: Fixed a bug in `action_iso.c` where the ISO filename and Volume ID were ignored due to a mismatch between JSON keys and C code.

"Code is poetry, but the changelog is the history."

blog

- https://penguins-eggs.net
- llms.txt